

ABDELLAH ELAZZAOU

AI Engineer | Machine Learning | Computer Vision | LLM & AI Agents

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PROFESSIONAL SUMMARY

AI Engineering student at ENSA Agadir specializing in building intelligent systems: LLM-powered agents, Retrieval-Augmented Generation (RAG), computer vision pipelines, and multi-modal AI assistants. Actively seeking AI internship opportunities.

WORK EXPERIENCE

AI & Computer Vision Intern Orange Summer Challenge 2025 Winner – Orange Morocco Jul – Sep 2025

- Designed a hybrid YOLO11 + CNN architecture for automated tomato disease detection and multi-class classification.
- Executed full ML lifecycle: data collection, image annotation with Roboflow, preprocessing, model training and evaluation.

Data Science Intern – CodeAlpha (Remote) Jun 2025

- Applied ML algorithms and EDA on real-world datasets; built end-to-end analysis pipelines using Python and Scikit-Learn.

AI PROJECTS

AI Personal Assistant – Gmail / Calendar / Sheets

Multi-tool AI agent using LlamaIndex that manages Gmail, Google Calendar, and Google Sheets through natural language, with human-in-the-loop validation before taking actions.

Stack: LlamaIndex, Python, Google APIs, AI Agents, Agentic Workflows

Health Assistant Chatbot – RAG System

Conversational health assistant leveraging LangGraph and a RAG pipeline to retrieve and synthesize accurate medical information from a curated knowledge base.

Stack: LangGraph, RAG, Python, Vector Database, LLMs

Construction Site PPE Safety Detection

Real-time computer vision system that detects whether workers are wearing required PPE and triggers automated alerts when safety violations are identified.

Stack: YOLOv11, OpenCV, Python, Object Detection, Computer Vision, Django, DRF, react.js

Vehicle & License Plate Recognition + Dwell Timer

Automated surveillance system detecting vehicles, extracting license plate text via OCR, and tracking dwell time in front of a camera. Applicable to parking and traffic management.

Stack: YOLO11, Pytesseract, OpenCV, OCR, Python

Gym Management System + AI Chatbot

Full-stack gym management platform and an integrated AI chatbot for automated support and queries, powered by LlamaIndex.

Stack: Django, DRF, React.js, LlamaIndex, PostgreSQL

EDUCATION

National School of Applied Sciences (ENSA), Agadir – Engineering – Data Science, Big Data & AI 2022 – 2027

ALX Program – Data Engineering & AI Certification 2024 – 2025

Baccalauréat – Sciences Physiques, Option Français – Mathematics & Physics 2022

TECHNICAL SKILLS

AI & ML: PyTorch, Scikit-Learn, CNN, YOLO (v8/v11), Transfer Learning, Model Fine-Tuning, Deep Learning

LLM & Agents: LlamaIndex, LangChain, LangGraph, RAG Systems, AI Agents, Prompt Engineering, Vector Databases

Development: Python, Django / DRF, React.js, Git, REST APIs

Cloud & Tools: Docker, GitHub, MLOps

CERTIFICATIONS

Data Science with Python (DataCamp, Feb 2025) | **Data Engineer Associate** (DataCamp, Jan 2025) | **ALX Data Analytics** (ALX, Sep 2025)

LANGUAGES

Arabic – Native

French – Intermediate

English – Intermediate